

EuroSAT Multispectral Dataset – EDA Report

Introduction

This report summarizes the exploratory data analysis (**EDA**) performed on the EuroSAT multispectral dataset. The dataset consists of satellite images organized into **10** land-use classes (**AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake**), each image represented with **13** spectral bands such as (**Blue (B02), Green (B03), Red (B04), Near Infrared (B08)**). The goal of EDA is to investigate dataset structure, spectral band behavior, vegetation patterns (NDVI), and feature separability to support classification tasks.

Dataset Structure

- **Size:** **27,000** images, each with **64×64** spatial resolution and **13** spectral bands.
- **Classes:** **10** distinct land-use categories stored in separate folders, enabling straightforward supervised classification.
- **Balance:** Moderately imbalanced distribution; most classes contain ~**3000** samples, while some (e.g., Pasture) are slightly underrepresented. Although the dataset is relatively balanced, slight **class imbalance** may **bias the model toward majority classes**. However, the imbalance is not severe enough to require heavy correction techniques.
- **Data Quality:** **No missing values** detected; images are complete and ready for analysis.

Spectral Band Behavior

- **Band Statistics:** Spectral bands exhibit different ranges and distributions, confirming variability across wavelengths.
- **Correlation Heatmaps:**
 - **Residential Insights**
 - ❖ **Strong positive correlations** appear among **mid-to-high** bands ($\approx$ bands **2–7** and **10–12**), suggesting redundancy in spectral information for built-up areas.
 - ❖ Lower bands (0–1) show **weaker** correlation with **higher bands**, indicating they may capture different surface properties (e.g., atmospheric or brightness effects).
 - ❖ Band **8–9** region shows **moderate** independence, which could help in distinguishing materials within residential zones.
 - **Forest Insights**
 - ❖ **Higher variability** in correlations compared to residential areas, indicating richer spectral diversity in vegetation.
 - ❖ Some **lower bands (0–1)** show **negative to weak** correlations with others, suggesting sensitivity to different vegetation or moisture characteristics.
 - ❖ **Strong correlations** among bands **4–7** and **10–12** imply these bands capture similar

vegetation signals (e.g., canopy structure or chlorophyll content).

- ❖ Band **8–9** again shows **moderate** independence, useful for vegetation discrimination.

- **Spectral Signatures:**

- **Forest** class shows **low reflectance in red bands** and **high reflectance in NIR**, typical of dense vegetation.
 - Variations across bands highlight unique spectral information useful for classification.
- **Scatter Plots:** Adjacent bands (e.g., **Band 2 vs Band 3**) show **strong positive correlation**, confirming redundancy but also variability for feature discrimination.

Vegetation Analysis (NDVI)

- **Average NDVI per Class:**

- **Forest** and **Pasture** have the **highest NDVI values (~0.7)**, reflecting dense and healthy vegetation.
 - **Water bodies** (SeaLake, River) and **urban classes** (Residential, Industrial, Highway) show **low or negative NDVI**.

- **NDVI Distribution:**

- Dataset-wide distribution **centers around 0.3–0.4**, indicating moderate vegetation density overall.
 - Class-wise analysis shows clear separation: **Forest** values cluster around **0.7–0.8**, while **Residential** peaks near **0.4**.

- **Insights:** NDVI is a strong discriminative feature for distinguishing vegetated vs. non-vegetated regions.

Feature Separability

- **PCA Visualization:** Reveals partial separability between classes in reduced feature space, with **overlaps among vegetation types**.
- **t-SNE Visualization:** Shows **clearer clustering** patterns compared to PCA, emphasizing **non-linear** separability and stronger local structure.

Visual Composites & Samples

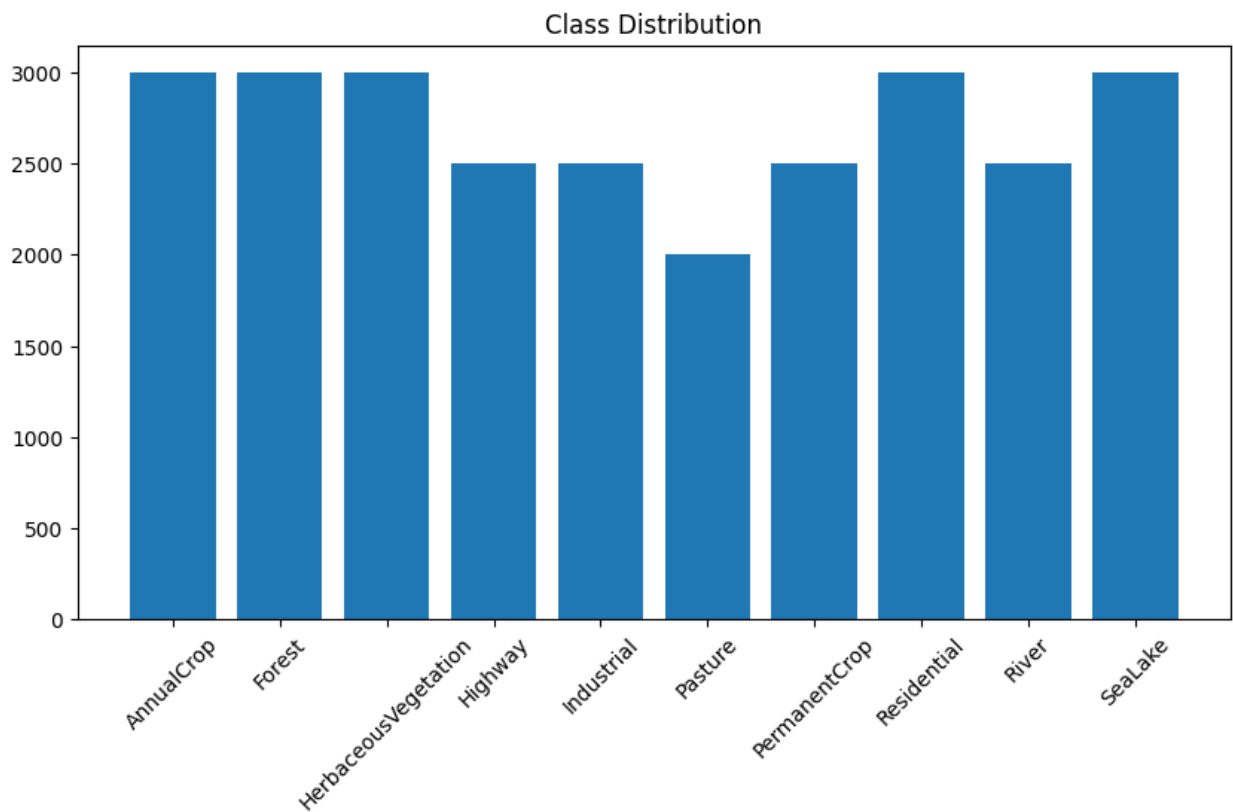
- Average RGB Composites: Provide muted but **representative visual summaries of each class**, reflecting **mean spectral values** rather than **natural colors**.
- Sample Images per Class: Show **intra-class consistency** (e.g., AnnualCrop fields) and **inter-class variability** (e.g., Forest vs Residential), confirming dataset suitability for supervised learning.

Key Insights

- Vegetation classes (Forest, Pasture, Crops) are spectrally distinct due to **strong NIR** reflectance and **high NDVI**.
- Urban classes show **weaker correlations** and **lower NDVI**, reflecting mixed surfaces.
- Water bodies are easily separable due to **low/negative NDVI** and distinct spectral signatures.

- Dimensionality reduction confirms meaningful structure in spectral data, supporting classification feasibility.
- Combining spectral analysis, NDVI, and dimensionality reduction provides a **comprehensive understanding of class differences**.

Visualizations

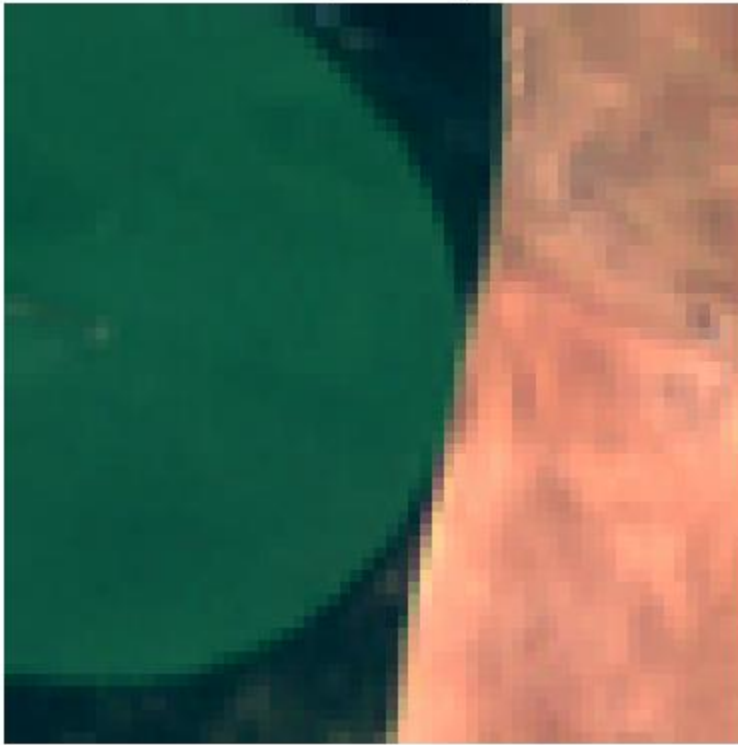


Class Distribution Visualization

This bar chart illustrates the distribution of samples across different land cover classes in the dataset. Most classes (such as AnnualCrop, Forest, Residential, and Industrial) have around **3000 samples**, while Pasture has **fewer (~2000)**. This visualization highlights the relative balance of the dataset and

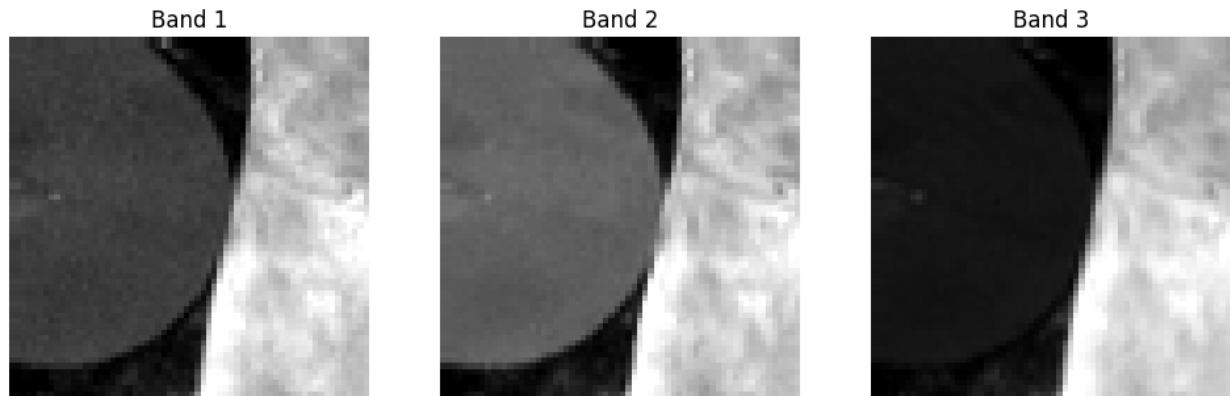
shows which classes are more represented, which is important for training and evaluating classification models.

Sample Image



Sample Image – Annual Crop

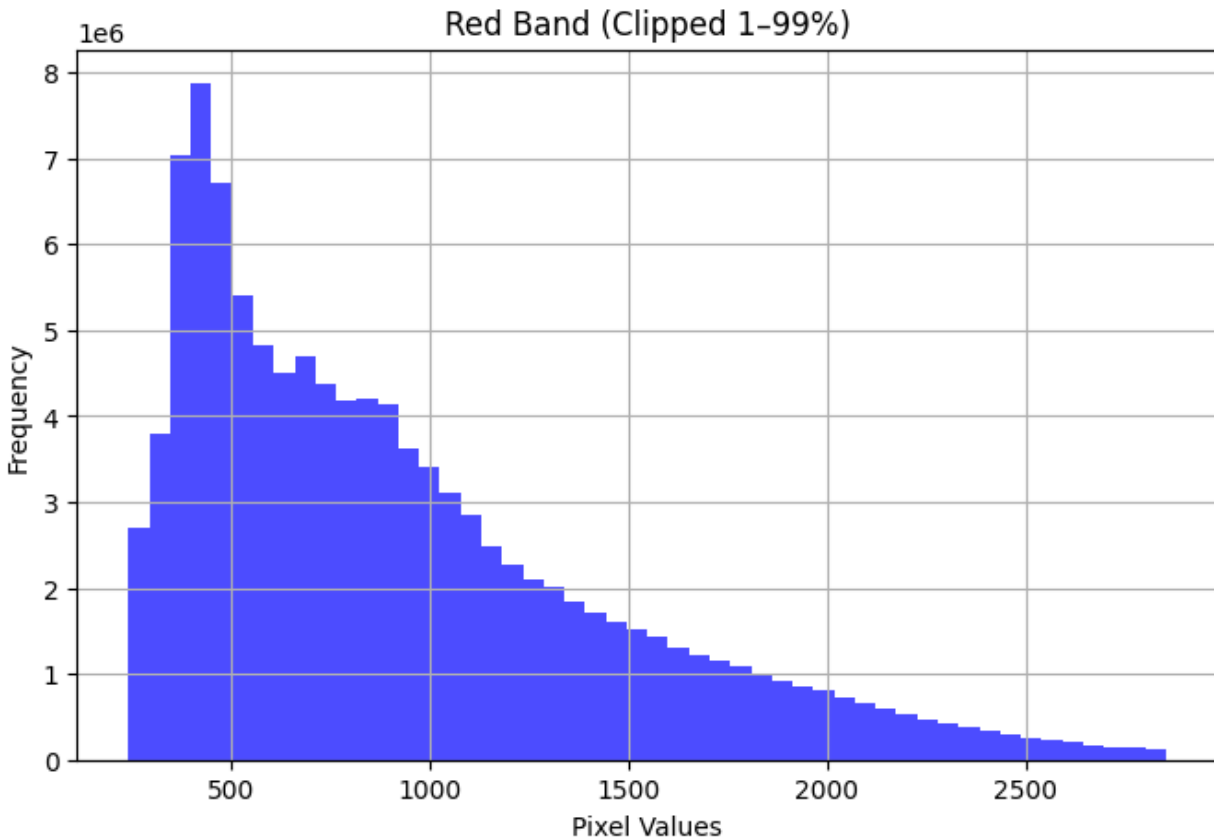
This sample image illustrates **annual crop** fields with clear irrigation patterns. The circular **green area** represents **actively cultivated crops**, while the adjacent **brown rectangular region** shows **dry soil or non-cultivated land**. The spectral distribution of annual crops is characterized by **high** reflectance in the near-infrared (**NIR**) **band** due to vegetation, **moderate** reflectance in the **red band**, and **lower** reflectance in the **blue band**. This contrast highlights the ability of remote sensing to distinguish between cultivated and uncultivated areas, making it useful for monitoring crop growth and land use.



Spectral Bands Visualization – Annual Crop

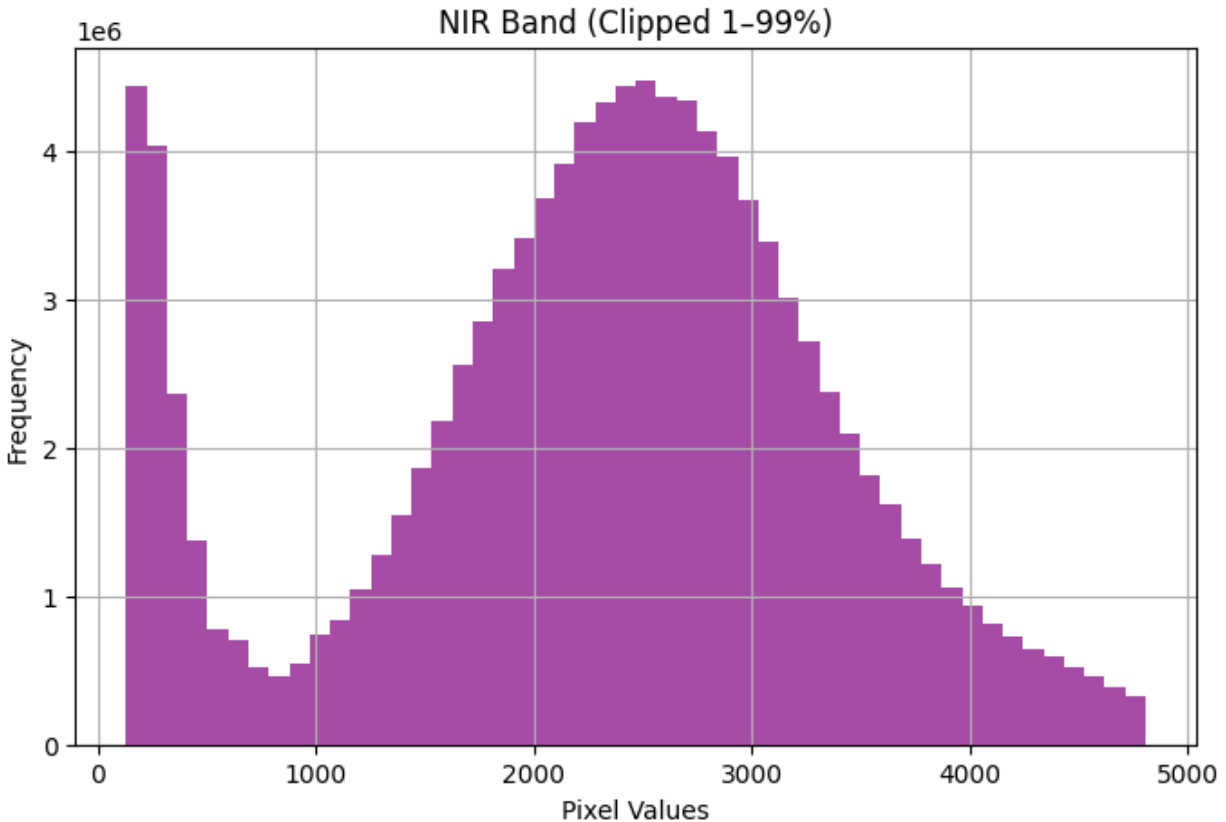
This visualization shows the individual spectral bands (Band 1, Band 2, Band 3) for the annual crop sample image. Each band highlights different reflectance properties of the land:

- **Band 1 (Blue region):** Sensitive to soil and water, showing **lower** reflectance for **vegetation**.
- **Band 2 (Green region):** Useful for assessing plant vigor, with **moderate** reflectance for **crops**.
- **Band 3 (Red region):** Strongly absorbed by chlorophyll, making **vegetation** appear **darker** compared to **soil**.



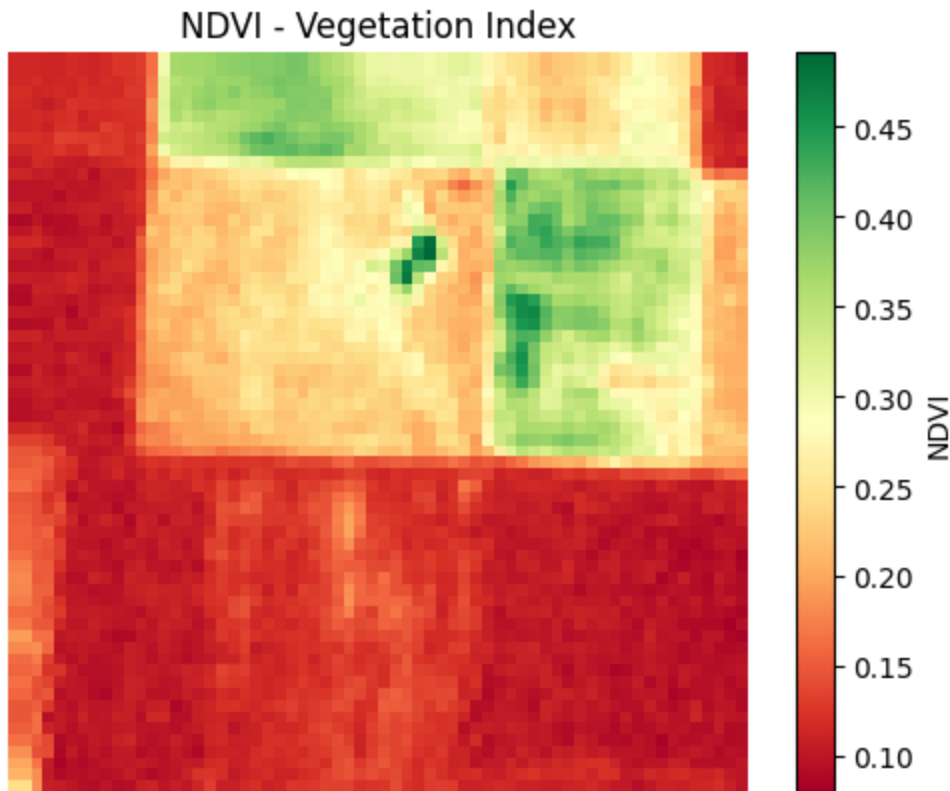
Spectral Distribution – Red Band

This histogram shows the distribution of pixel values in the **red spectral band** after clipping **extreme values (1–99%)**. The distribution is **right-skewed**, with a peak around mid-range values and a gradual decline toward higher reflectance. This pattern reflects how vegetation absorbs red light due to chlorophyll, resulting in lower pixel values for crop areas compared to soil or non-vegetated regions. Such histograms are useful for analyzing brightness and contrast, and for distinguishing vegetation from other land cover types.



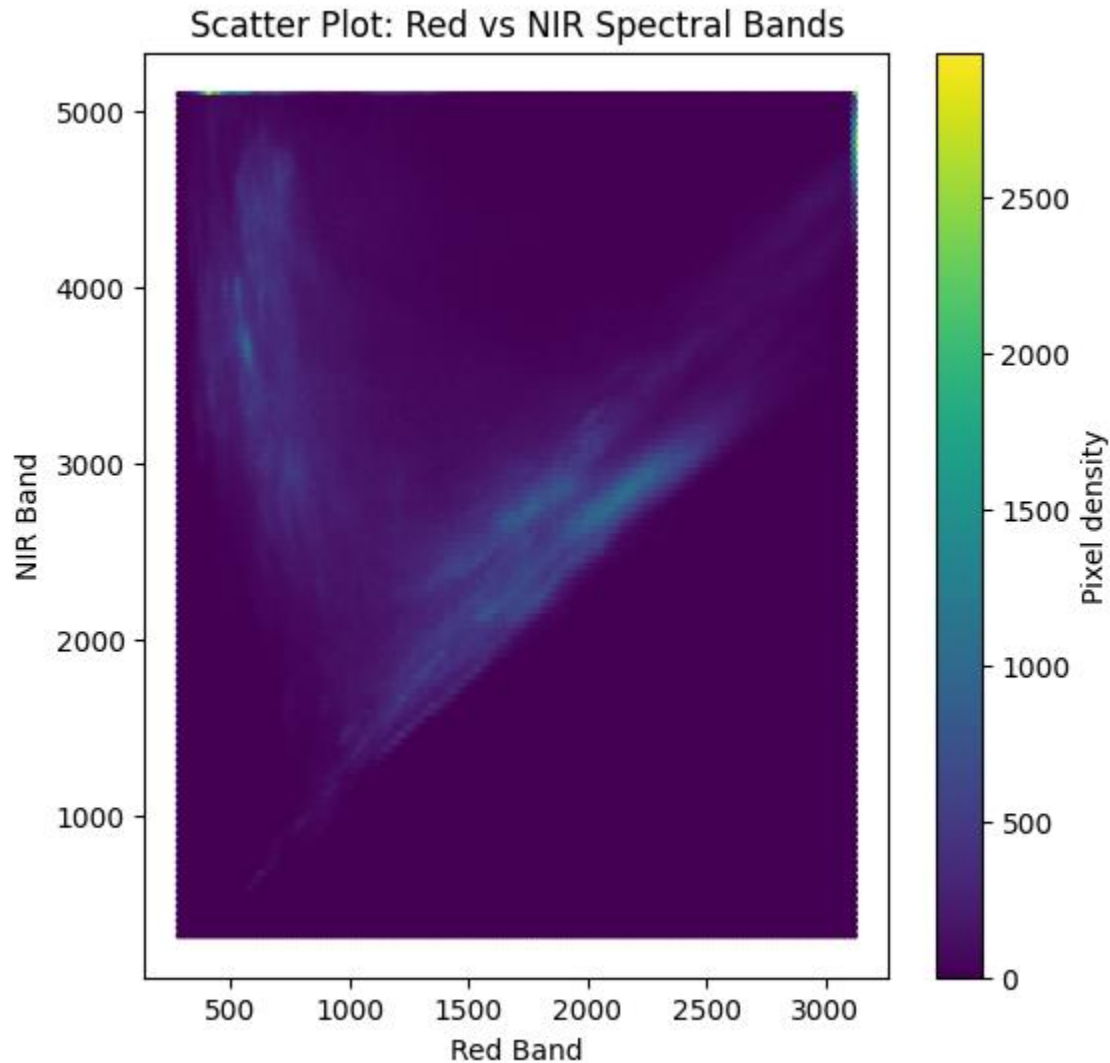
Spectral Distribution – NIR Band

This histogram shows the distribution of pixel values in the near-infrared (**NIR**) band after clipping **extreme values (1–99%)**. The distribution reveals a broad **peak** around **mid-to-high reflectance values**, which is typical for vegetation. Healthy crops strongly reflect NIR radiation due to their cellular structure, resulting in higher pixel values compared to soil or water. This spectral property is crucial for distinguishing vegetation from other land cover types and for monitoring crop health and growth patterns.



NDVI – Vegetation Index

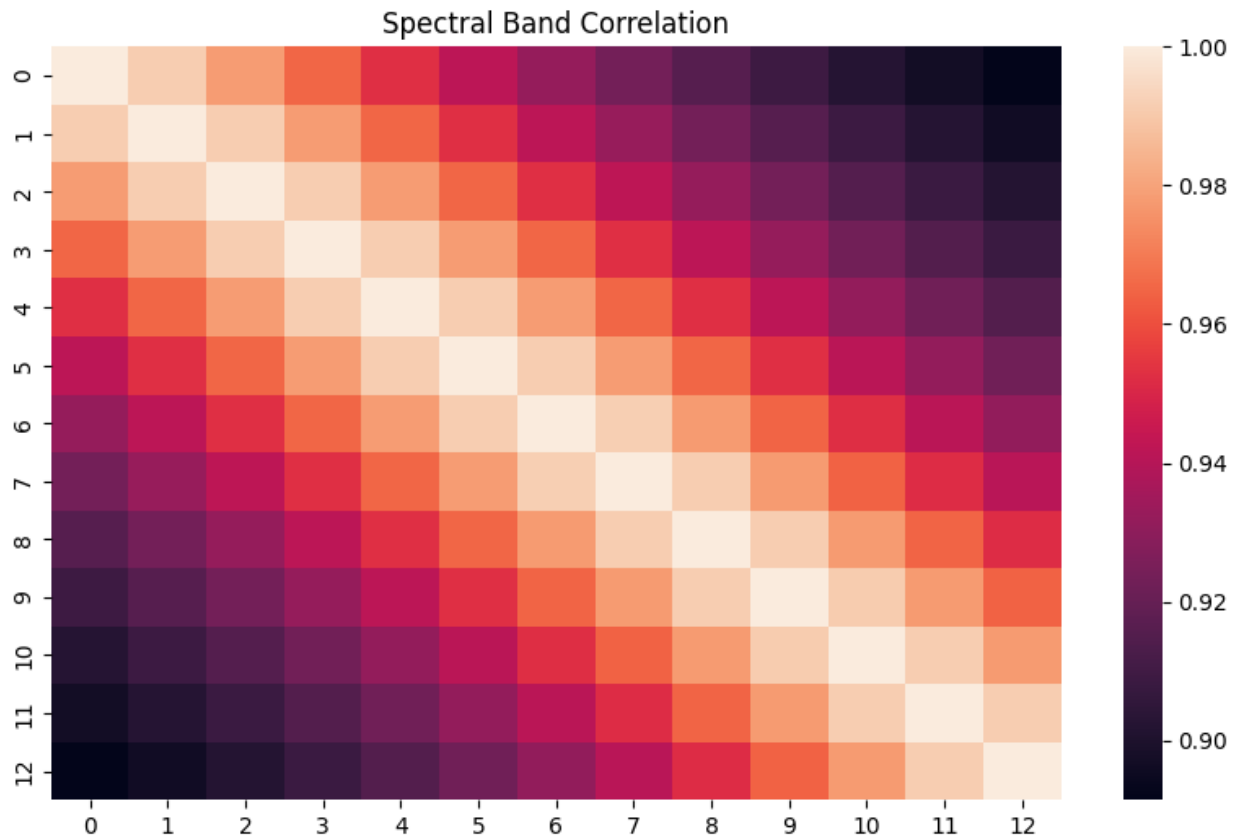
This visualization shows the **NDVI map** derived from the **red and near-infrared (NIR) bands**. NDVI values range from low (red areas, indicating barren land or sparse vegetation) to high (green areas, representing dense and healthy vegetation). The index highlights spatial variations in crop health and vegetation density, making it a powerful tool for agricultural monitoring, land management, and environmental studies. Areas with higher NDVI values correspond to vigorous plant growth, while lower values indicate stressed or non-vegetated regions.



Scatter Plot – Red vs NIR Spectral Bands

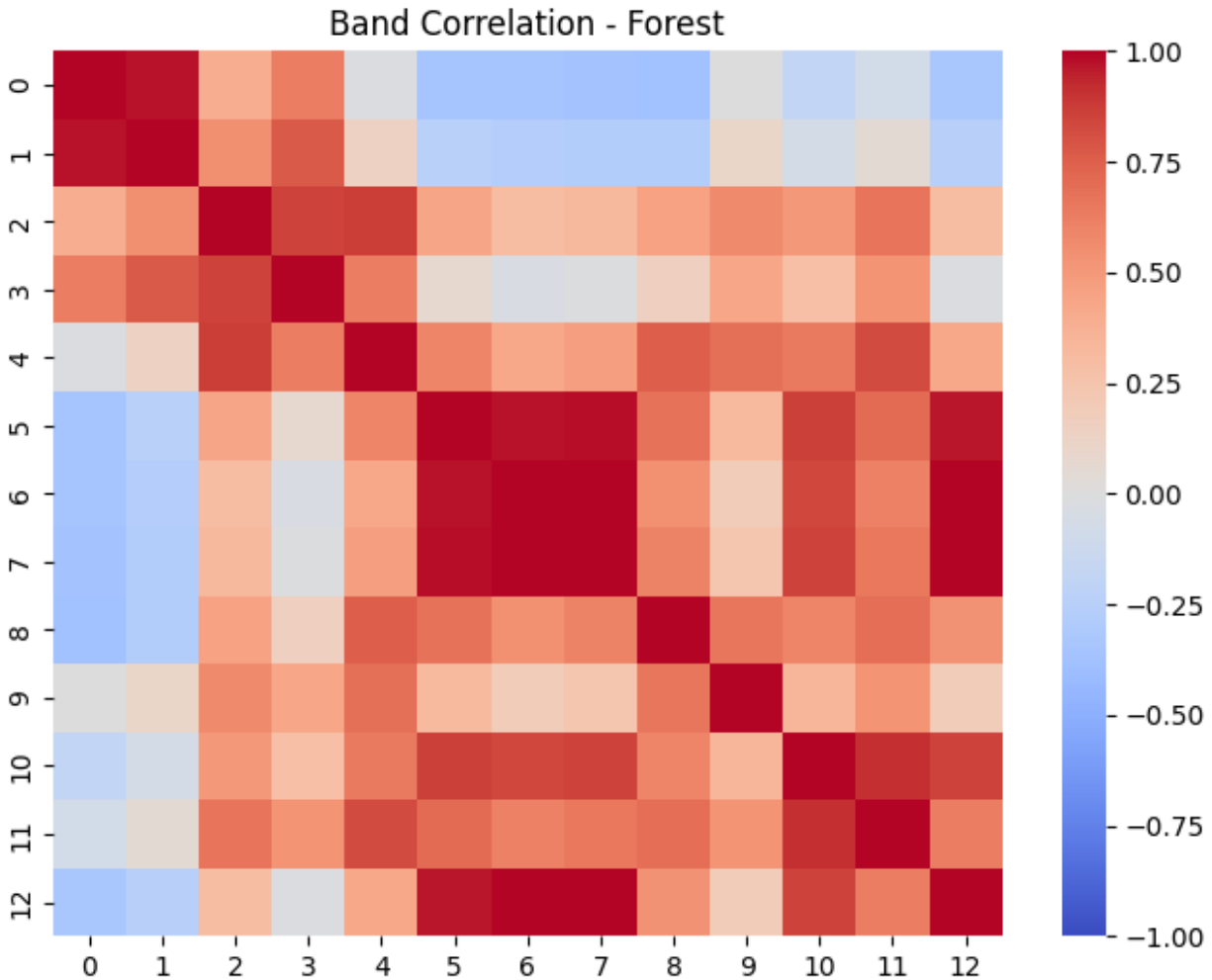
This **scatter plot** visualizes the relationship between the **red** and **near-infrared (NIR) spectral bands** for a sample of pixels. Vegetation typically shows **low** reflectance in the **red band** (due to chlorophyll absorption) and **high** reflectance in the **NIR band** (due to leaf structure), which creates a clear separation in the plot. The dense cluster of points highlights the distribution of vegetation pixels, while non-vegetated areas (such as soil or water) appear in different regions of the plot.

This visualization is crucial for understanding spectral signatures and for developing vegetation indices such as NDVI.



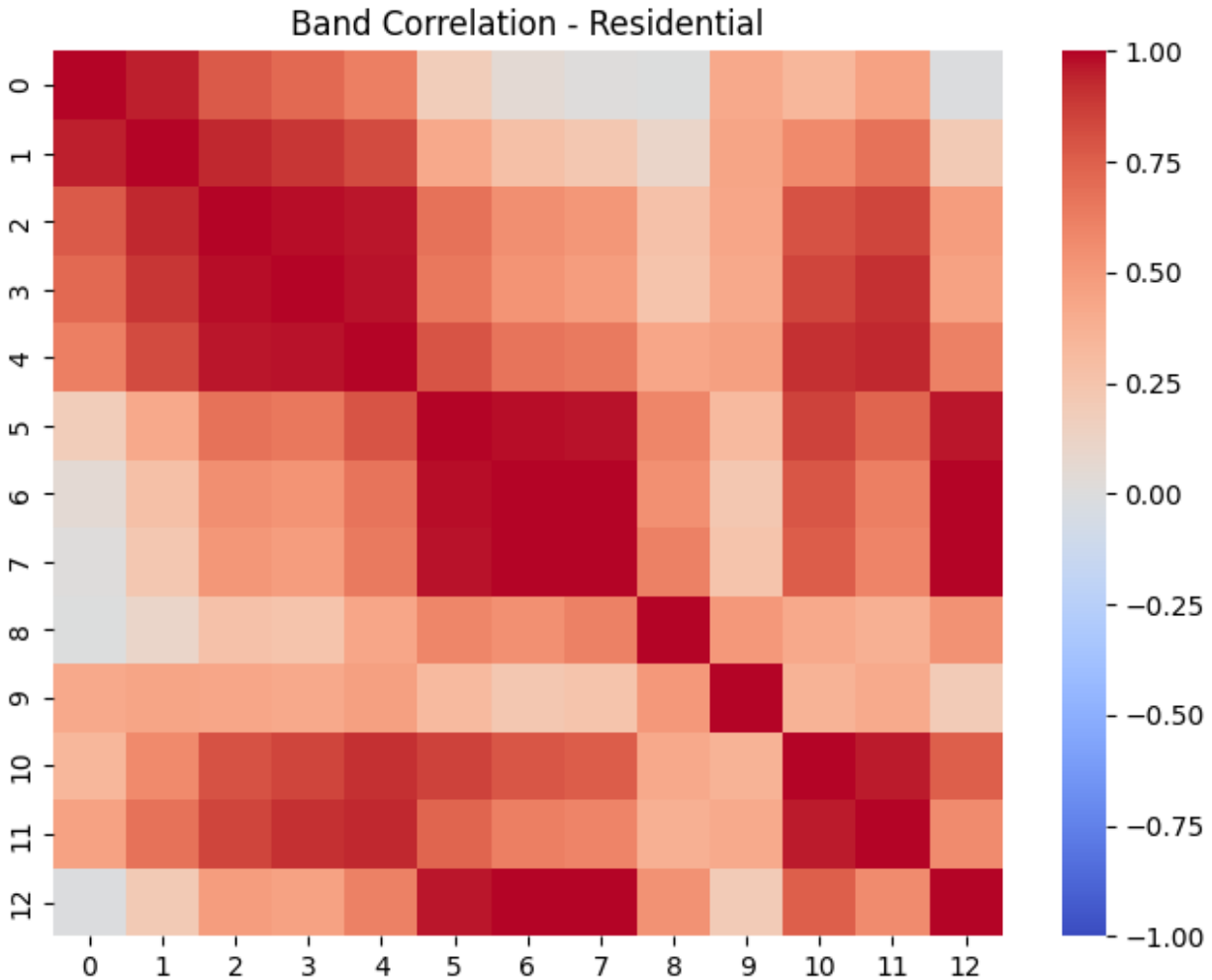
Spectral Band Correlation

This **heatmap** shows the **correlation values between spectral bands in the dataset**. The **diagonal** highlights perfect correlation (**value = 1**) within each band, while **adjacent bands** show **strong correlations** due to overlapping spectral information. **Distant bands** exhibit **lower correlation**, reflecting their ability to capture distinct spectral properties. This visualization is important for feature selection and dimensionality reduction, as it helps identify redundant bands and highlights those that provide unique information for land cover classification.



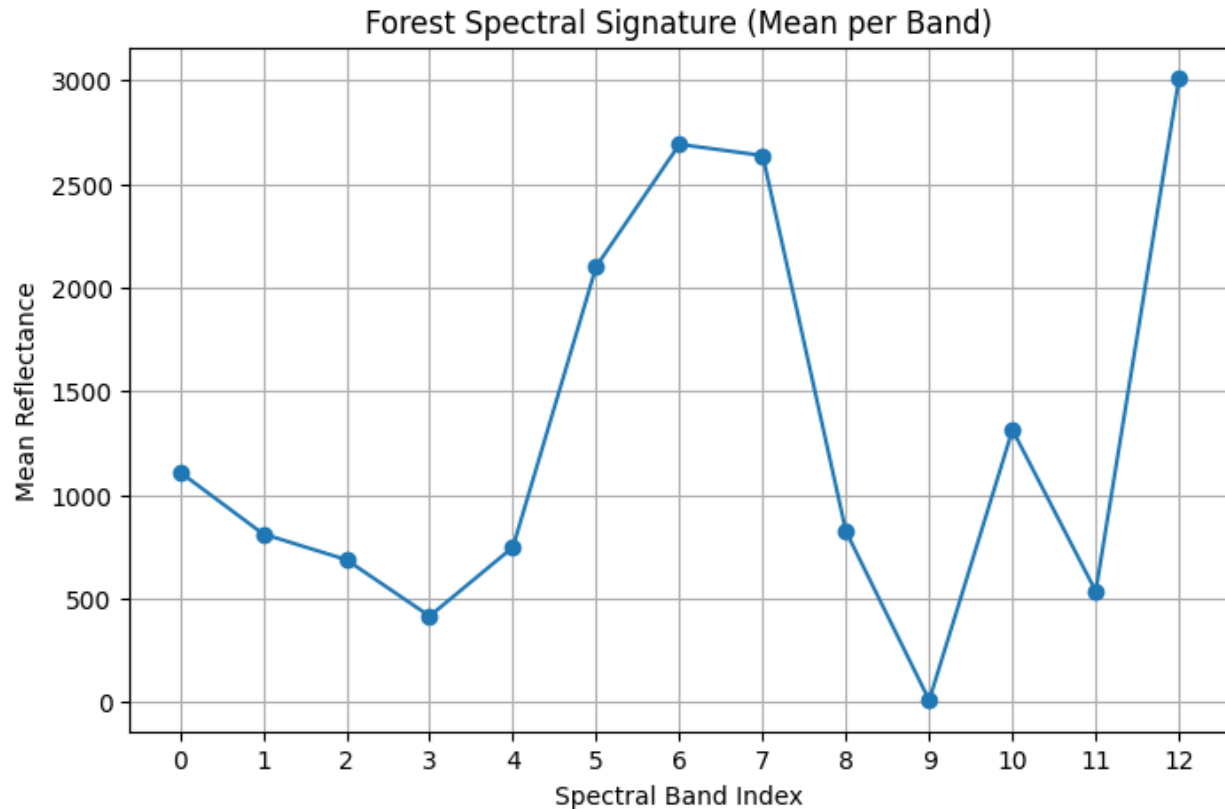
Band Correlation – Forest

This **heatmap** shows the **correlation between spectral bands for forest samples**. Most bands exhibit **strong positive correlations**, reflecting the consistent spectral behavior of dense vegetation. The strong similarity across bands indicates redundancy, but also highlights the stability of forest spectral signatures. **Negative correlations** in some distant bands suggest **unique spectral information** that can be leveraged for classification.



Band Correlation – Residential

This **heatmap** illustrates the **correlation between spectral bands for residential areas**. Compared to forest, residential samples show **weaker and more variable correlations**, reflecting the heterogeneous nature of urban surfaces (concrete, asphalt, rooftops, vegetation patches). The mixed spectral signatures result in both strong and weak correlations, making urban classification more complex and requiring careful band selection.

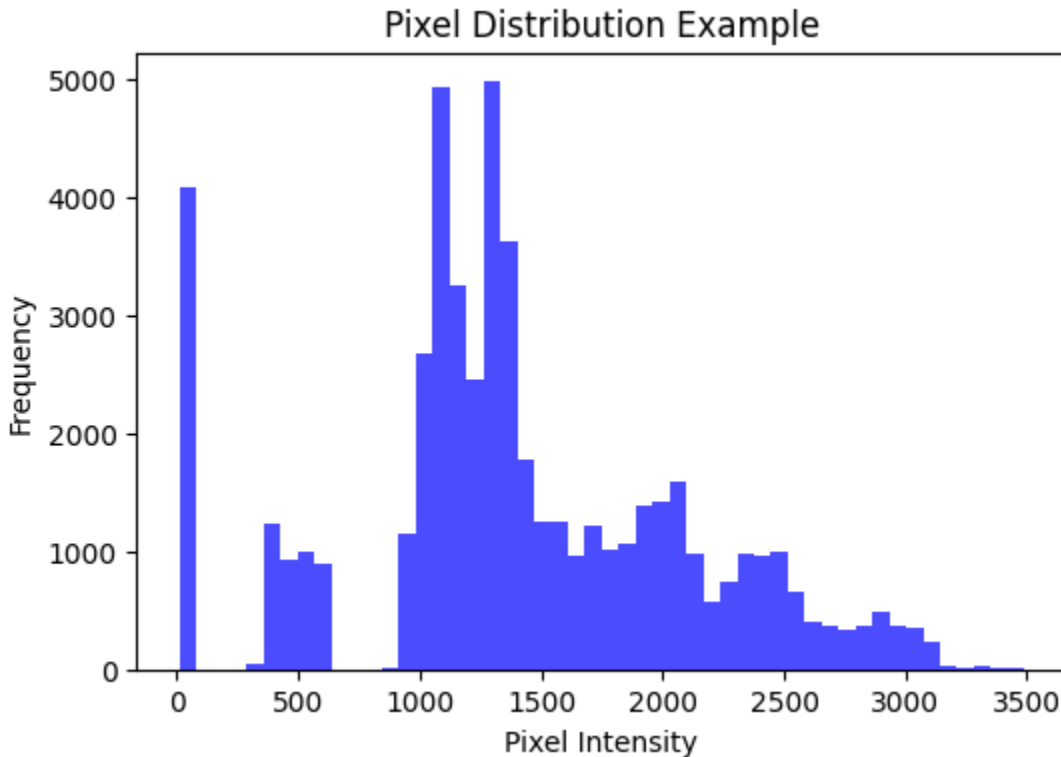


Forest Spectral Signature (Mean per Band)

This line graph shows the **mean reflectance values** across **spectral bands for forest samples**. The spectral signature of forests is characterized by:

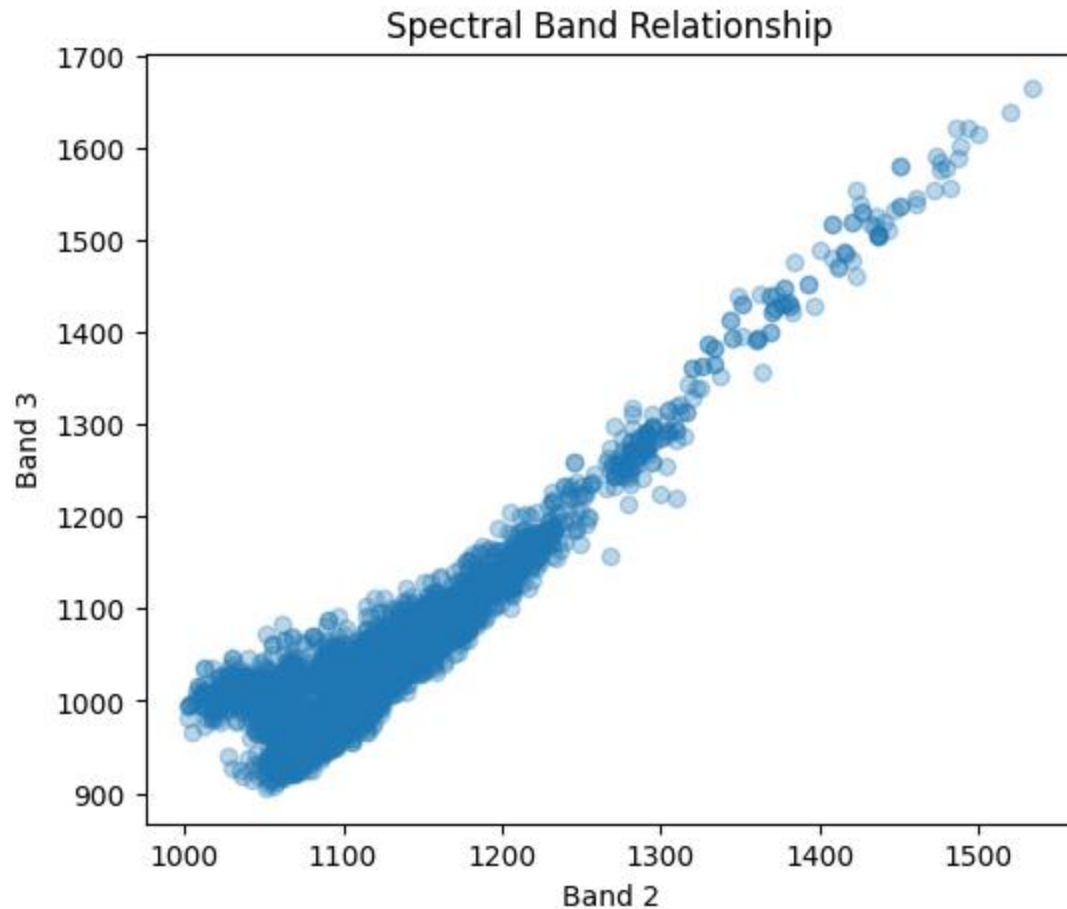
- **Low** reflectance in the **red band** due to strong chlorophyll absorption.
- **High** reflectance in the near-infrared (**NIR**) **bands**, which is typical of dense vegetation because of leaf structure.
- **Distinct peaks** and troughs across **other bands**, reflecting the complex interaction of forest canopy with light.

This spectral pattern is crucial for distinguishing forested areas from other land cover types and for monitoring vegetation health and density.



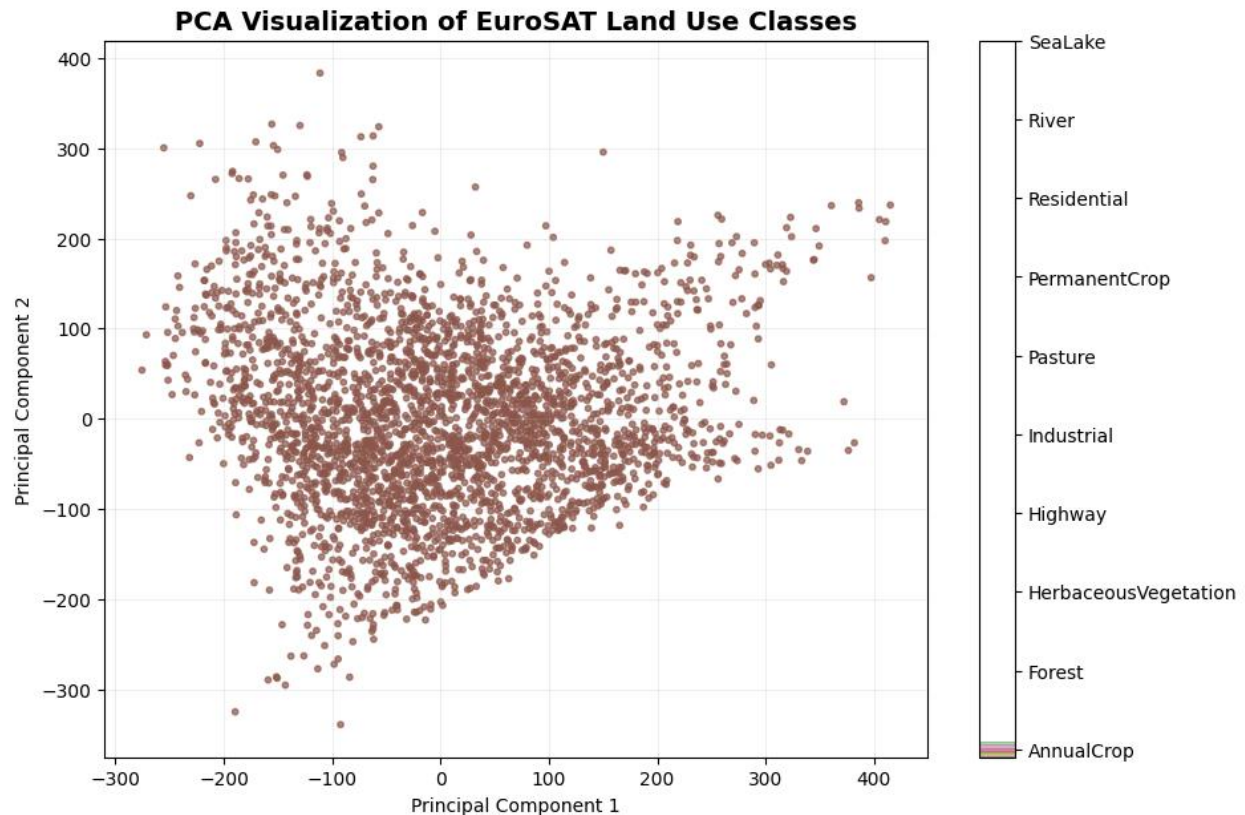
Pixel Distribution Example

This **histogram** shows the **distribution of pixel intensity** values for the sample image. Peaks at specific intensity ranges indicate common reflectance levels across the dataset. For example, lower intensities often correspond to shaded or water-covered areas, while mid-range values represent vegetation, and higher intensities may reflect bright surfaces such as soil or urban structures. Analyzing **pixel distributions** helps in understanding **image brightness, contrast, and the spectral variability** of different land cover types.



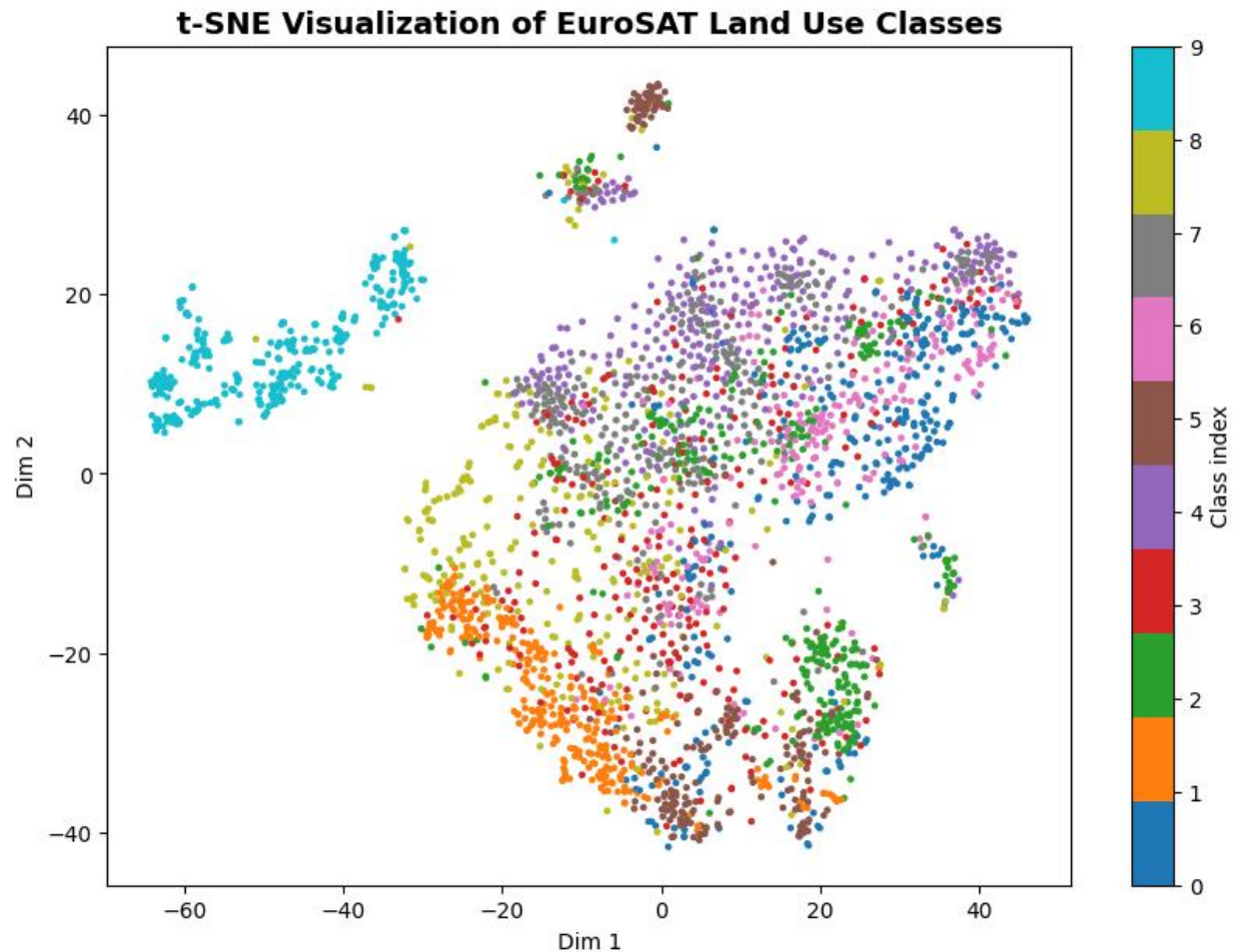
Spectral Band Relationship

This **scatter plot** illustrates the relationship between **Band 2** and **Band 3** pixel values. The clear **positive** correlation indicates that as reflectance in **Band 2 increases**, reflectance in **Band 3 also increases**. Such correlations are common between adjacent spectral bands due to overlapping wavelength ranges. This visualization helps in understanding redundancy among bands and highlights the importance of selecting distinct spectral regions for effective land cover classification.



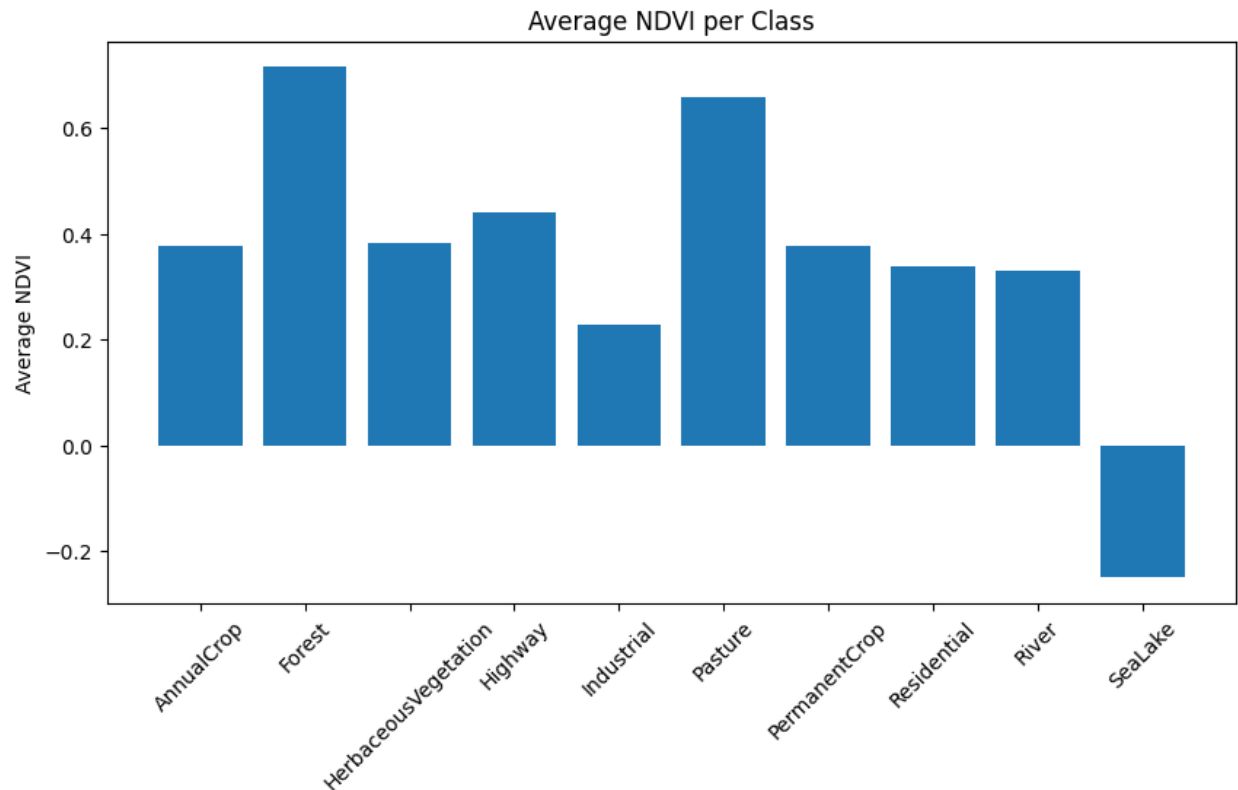
PCA Visualization of EuroSAT Land Use Classes

The **PCA** plot shows significant **overlap between classes**, indicating poor separability in 2D space. This suggests that PCA does not capture enough discriminative features for this dataset. More **advanced methods** may be needed for better class separation.



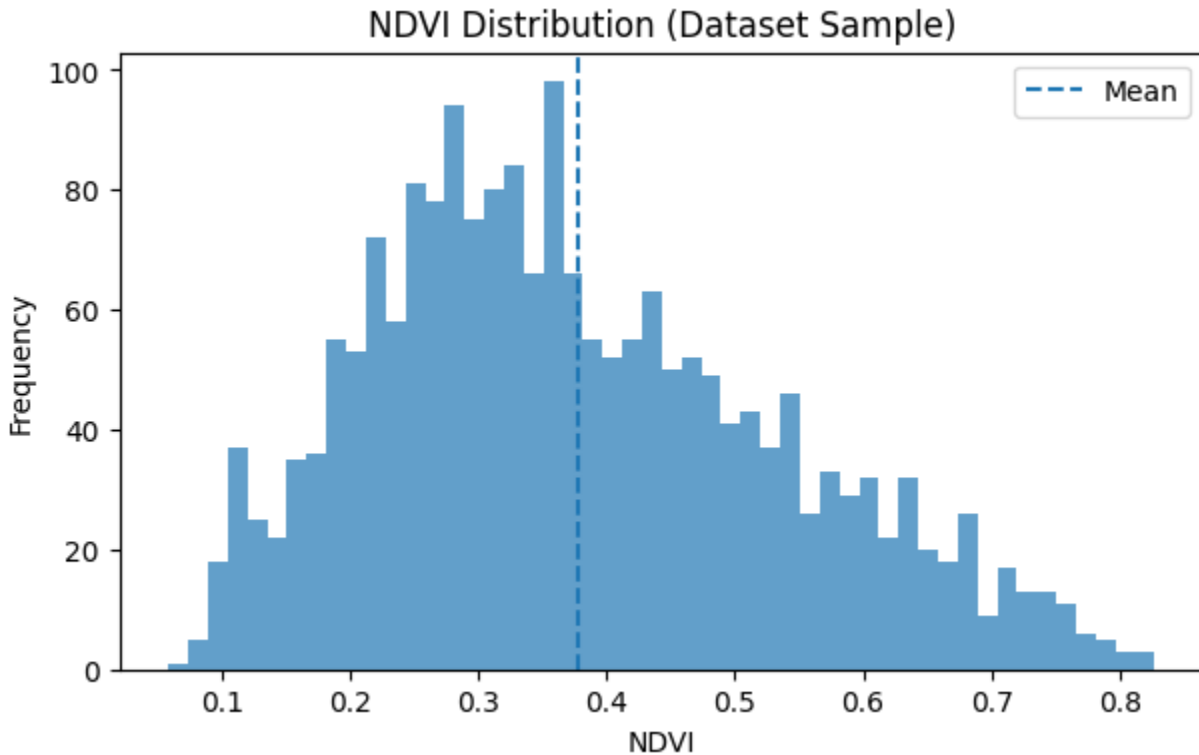
t-SNE Visualization of EuroSAT Land Use Classes

This **scatter plot** shows the distribution of EuroSAT land use classes using **t-SNE dimensionality reduction**. Each point represents a sample image projected into **two dimensions**, with colors indicating different land cover categories (such as AnnualCrop, Forest, Residential, River, and SeaLake). t-SNE emphasizes local structure, revealing clusters of similar classes and highlighting overlaps where spectral signatures are less distinct. This visualization is useful for **exploring class separability, identifying challenging overlaps, and gaining insights** into the spectral diversity of land use categories.



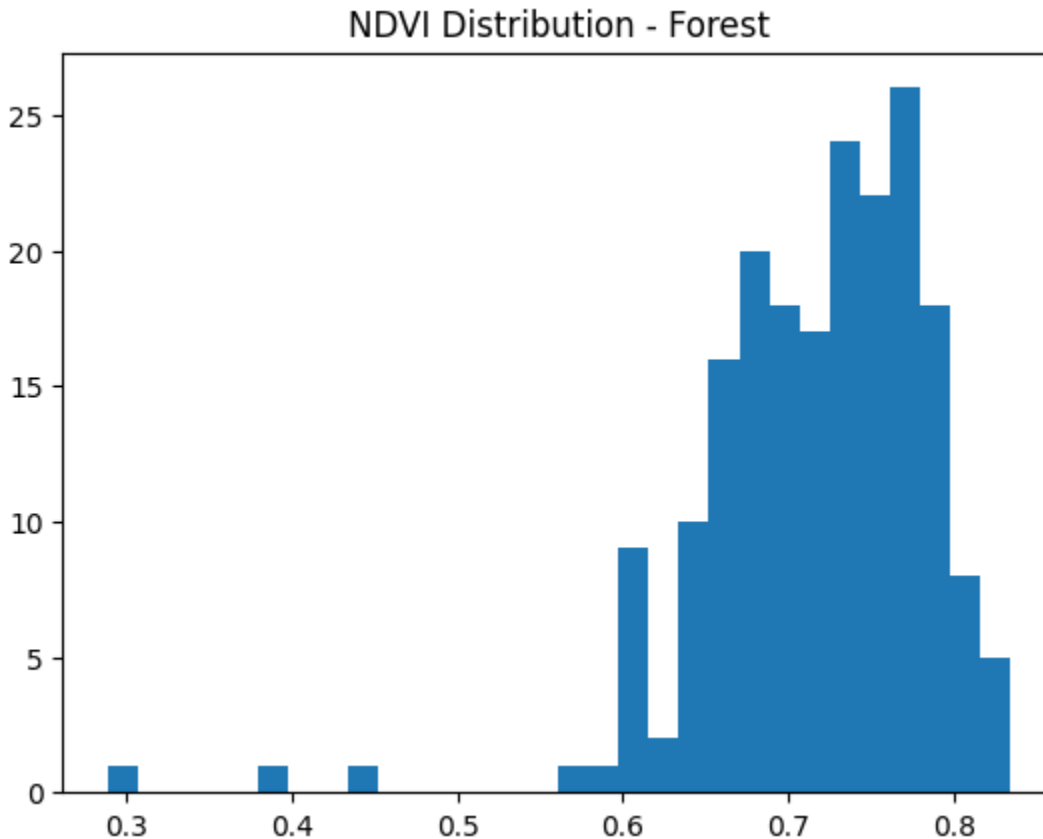
Average NDVI per Class

This **bar chart** compares the **average NDVI** values across different land use classes. **Forest and Pasture** show the **highest NDVI** values, reflecting dense and healthy vegetation. **PermanentCrop and HerbaceousVegetation** also exhibit relatively **high NDVI**, consistent with active plant growth. In contrast, classes such as **Residential, Industrial, Highway, River, and SeaLake** have **low or negative NDVI** values, indicating sparse vegetation or non-vegetated surfaces. This visualization highlights the effectiveness of NDVI in distinguishing vegetated areas from urban, water, or barren land cover types.



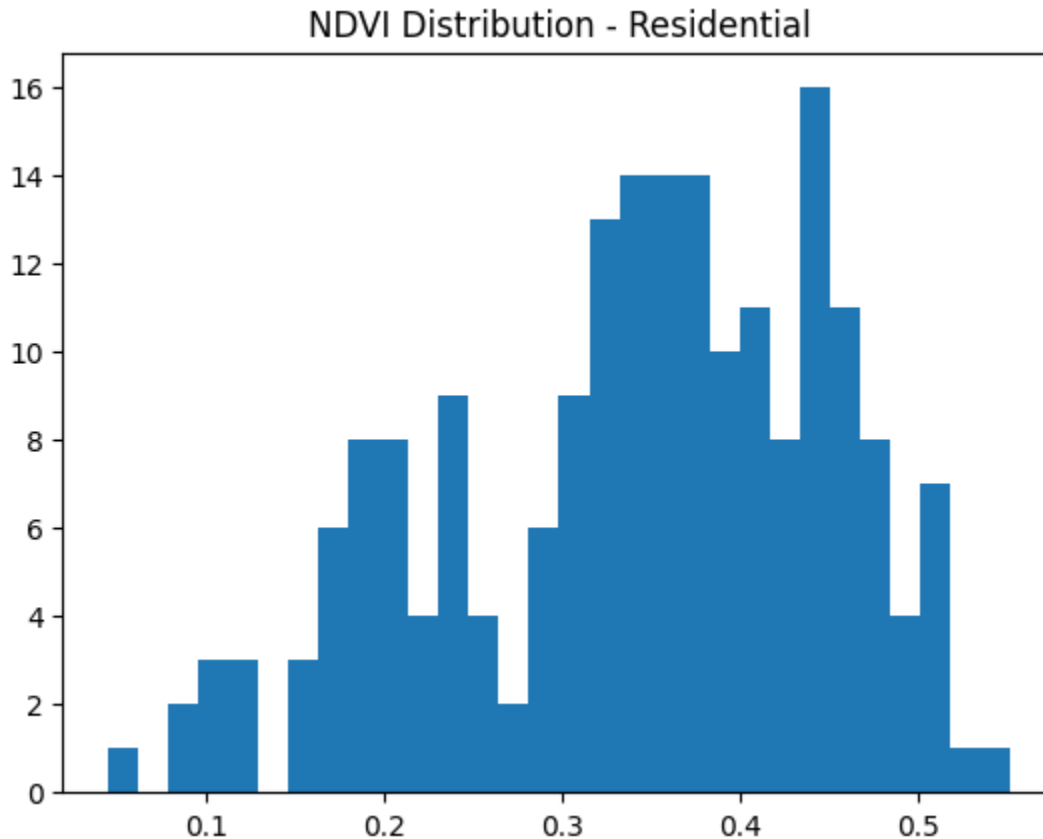
NDVI Distribution (Dataset Sample)

This **histogram** shows the distribution of **NDVI values** across a dataset sample. Most values **cluster around 0.3–0.4**, reflecting moderate vegetation density, while fewer samples exhibit very low or very high NDVI. The **dashed line** indicates the **mean NDVI**, which provides a useful benchmark for overall vegetation health in the dataset. The bell-shaped but slightly skewed distribution highlights the variability of land cover, with dense vegetation (higher NDVI) and non-vegetated surfaces (lower NDVI) both represented. This analysis is essential for understanding the general vegetation condition across the study area.



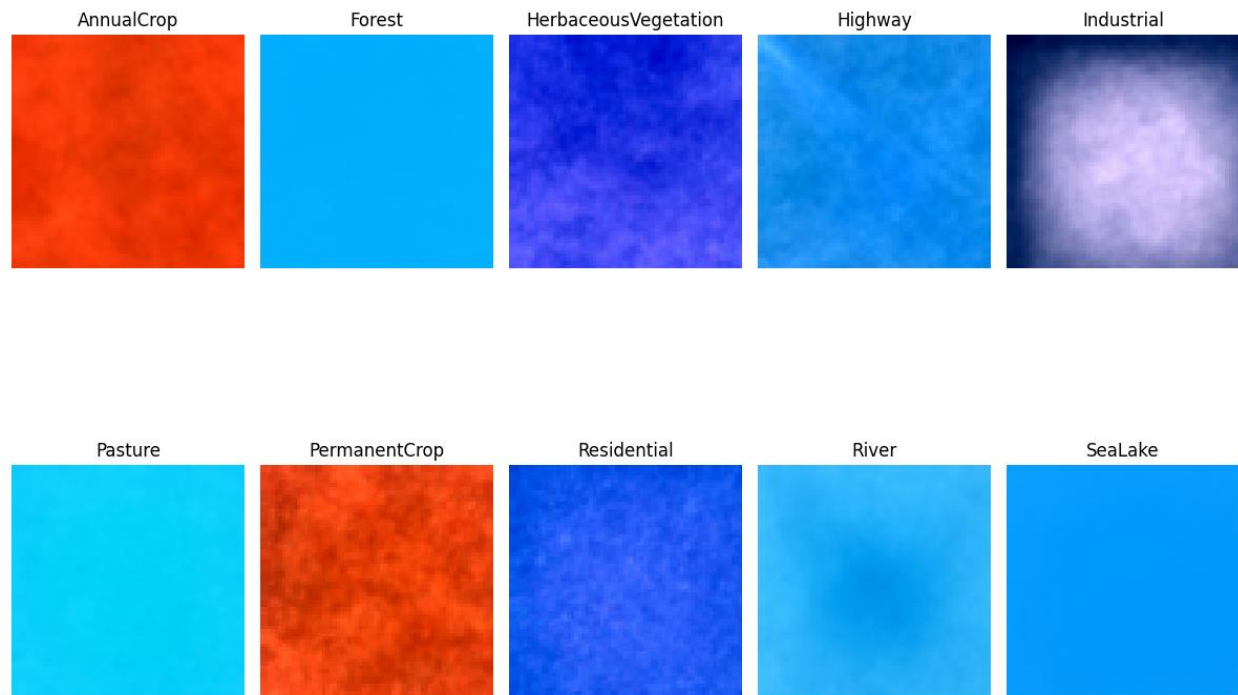
NDVI Distribution – Forest

This **histogram** shows the distribution of **NDVI values for forest samples**. Most values fall **between 0.6 and 0.8**, with a **peak around 0.75–0.8**, reflecting dense and healthy vegetation. The high NDVI values are consistent with strong chlorophyll absorption in the red band and high reflectance in the NIR band, typical of forested areas. This distribution highlights the stability and vigor of vegetation in forest regions.



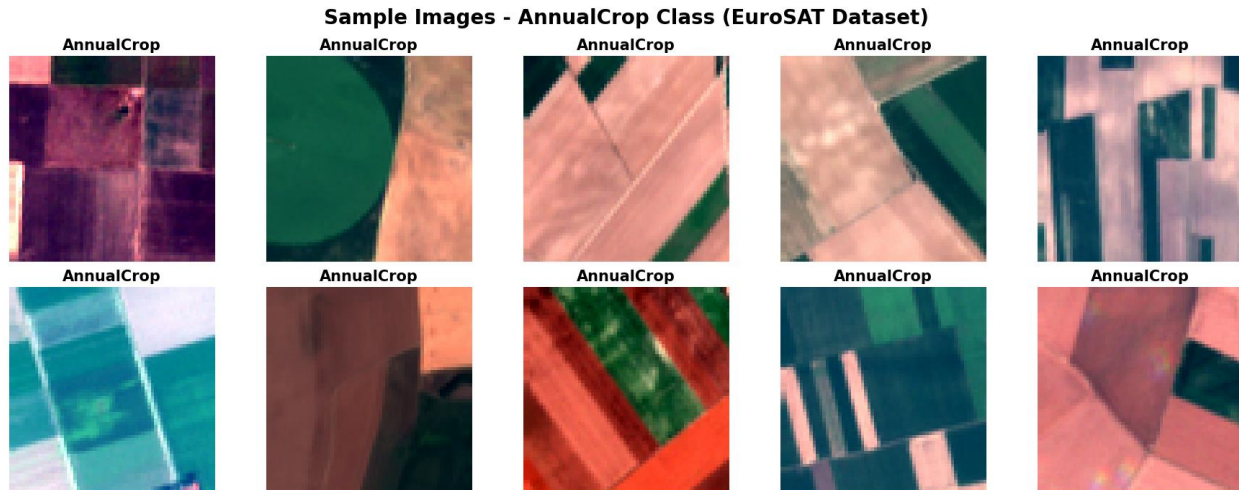
NDVI Distribution – Residential

This **histogram** illustrates the **NDVI distribution** for **residential areas**. Most values **cluster between 0.3 and 0.45**, with a **peak around 0.4**, indicating moderate vegetation density. The lower NDVI values compared to forest reflect the mixed nature of urban environments, where vegetation is interspersed with non-vegetated surfaces such as buildings, roads, and rooftops. This distribution emphasizes the spectral variability and reduced vegetation cover in residential zones.



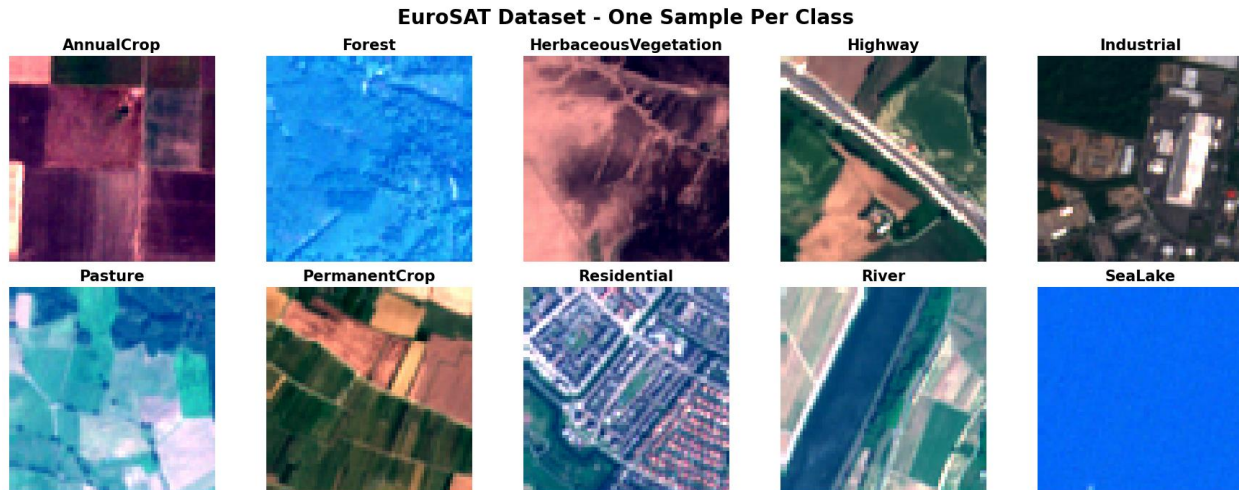
Average RGB Composite per Class

This **visualization** shows the **average RGB** composites for different land use classes. The **colors do not necessarily match the natural appearance of each class** (e.g., forests may not appear purely green, and water bodies may not appear purely blue). Instead, the **composites represent the mean spectral values across all samples in each category**. This averaging process blends variations within the class, producing muted or mixed colors that reflect the diversity of surfaces (vegetation, soil, infrastructure, or water) present in the dataset. The visualization is useful for comparing spectral differences between classes, even if the colors differ from their real-world appearance.



Sample Images – AnnualCrop Class (EuroSAT Dataset)

This panel displays **sample satellite images from the AnnualCrop class**. The variations in color and texture reflect different crop types, soil conditions, and growth stages. Geometric field patterns are clearly visible, highlighting the structured nature of agricultural landscapes. The diversity of spectral responses—from **green vegetation** to **brown soil** and **reddish tones**—illustrates the complexity of agricultural monitoring. These samples provide valuable insights into how annual crops appear in satellite imagery, supporting classification and land use analysis.



EuroSAT Dataset – One Sample Per Class

This panel presents **one representative sample image for each land use class** in the EuroSAT dataset. The diversity of colors and textures highlights the distinct spectral and spatial characteristics of different land cover types:

- **AnnualCrop / PermanentCrop / Pasture:** Agricultural fields with geometric patterns and varying vegetation stages.
- **Forest / HerbaceousVegetation:** Natural vegetation with dense or sparse coverage.
- **Residential / Industrial / Highway:** Built-up areas showing infrastructure, roads, and housing grids.
- **River / SeaLake:** Water bodies with varying reflectance depending on turbidity and surrounding land.

This visualization provides a clear overview of the dataset's scope, illustrating how satellite imagery captures the complexity of landscapes and why these classes are suitable for machine learning classification tasks.

Spectral Interpretation Summary

- The **spectral analysis** reveals that different land cover types exhibit **distinct reflectance** patterns **across spectral bands**. **Vegetation** classes show **low** reflectance in the **red band** and **high** reflectance in the **near-infrared (NIR) band** due to chlorophyll absorption and internal leaf structure.
- **Urban** areas exhibit **more uniform and less distinctive** spectral patterns due to mixed materials such as concrete and asphalt.
- **Water bodies** show consistently **low** reflectance across most bands, particularly in the **NIR region**.
- These **differences** confirm that multispectral data provides **strong discriminative features for land use classification tasks**.

Conclusion

- The **EuroSAT multispectral dataset** provides rich spectral and spatial information suitable for **land-use classification tasks**.
- **Exploratory Data Analysis** revealed clear **differences in spectral band behavior** across different land cover types.
- **NDVI** proved to be a strong feature **for distinguishing vegetation from non-vegetation areas**.
- Dimensionality reduction techniques (**PCA and t-SNE**) showed **meaningful class separability** in the feature space.
- Overall, the dataset demonstrates strong potential for accurate land type classification using deep learning models.